Multitask Recognition of Types and Operating States of Underwater Engines based on Mel Spectrogram Decomposition in a GRU-with-Attention Model

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Abstract: This work addresses *multitask* recognition of the **active engine** (M1–M5) and its **operating state** from underwater audio. We compare four shared feature-extraction networks, here termed *backbones*, namely BiLSTM+attention, GRU+attention, a temporal Transformer, and ResNet-50 on spectrograms, all coupled to conditional state heads. Preprocessing uses 0.5 s windows of 64-bin *log-mel* spectrograms, *z-score* normalization, and light augmentation (random gain, Gaussian noise, and *SpecAugment*). Experiments are conducted on the *single-engine* subset of *Wolfset*, with evaluation at segment and file levels. Among the reference models, GRU and Transformer reach file-level F1 of **1.00** for engine and up to **0.68** for state. Motivated by these results, we propose a **sub-spectrogram GRU** variant; with B=8, it yields the best trade-off (mean F1 = 0.800; file-F1: engine = 1.00, state = 0.74). Removing augmentation substantially degrades state recognition (file-F1 0.74→0.47). On a `Tesla T4` GPU, end-to-end inference over a complete file under the adopted windowing required **83–113 s** with memory usage < 225 **MB**, supporting batch or near-online monitoring rather than strict real-time deployment.

Keywords: Underwater acoustics; Multitask learning; Engine state recognition; LSTM/Transformer; Sub-spectrograms; Naval applications.

1. INTRODUCTION

Acoustic warfare has played a decisive role in modern naval warfare. During World War II, the Battle of the Atlantic consolidated the systematic use of underwater sensors for submarine detection and classification, including ASDIC, the forerunner of active sonar, and countermeasures such as the *Bold* decoy Hackmann (1984); Urick (1983). During the Cold War, the United States Navy's *Sound Surveillance System* (SOSUS) enabled large-scale acoustic surveillance and submarine tracking, providing informational advantage in strategic scenarios such as the Cuban Missile Crisis Whitman (2005); Etter (2018). These developments established acoustic warfare as a key domain for controlling the underwater spectrum.

According to U.S. Department of Defense (2021), *acoustic warfare* involves the use of underwater acoustic energy to exploit, deny, or protect access to the sound spectrum. It comprises three subdomains: *support measures* (AWSM), based on passive interception and analysis of radiated energy; *countermeasures* (AWCM), based on deliberate emissions to deceive or saturate sensors; and *counter-countermeasures* (AWCCM), intended to preserve friendly acoustic capabilities under interference.

Automatic identification of underwater engines from acoustic signatures is central to passive maritime monitoring, supporting inference on operating states and behavioral patterns of naval and unmanned platforms. This work addresses the simultaneous identification of the active engine and its operating state from broadband hydroacoustic recordings. The problem is formulated as *multitask learning*, in which a shared feature-extraction network, termed *backbone*, feeds an engine classifier and engine-conditioned state heads.

Underwater acoustic classification is affected by non-stationary noise, gain and range variability, multipath, spectral overlap among operating modes, and class imbalance. To mitigate these effects, we use normalized *log-mel* spectrograms, light data augmentation, including *SpecAugment*, controlled gain, and noise, and weighted losses based on *Focal Loss*.

The main contributions are: (i) a multitask architecture with conditional state heads per engine; (ii) a *log-mel* extraction pipeline with light augmentation; (iii) an evaluation protocol with file-based partitioning and segment- and file-level metrics; (iv) an ablation across BiLSTM, GRU, Transformer, and ResNet-50 backbones; and (v) a reproducible implementation with checkpoints and metric history. The paper is organized as follows: Sec. 2 reviews related work, Sec. 3 describes the methodology, Sec. 4 presents the results, and Sec. 5 concludes the paper.

2. RELATED WORK

2.1 Underwater Target and Engine Recognition

Automatic recognition of underwater acoustic sources has advanced considerably with the development of hydroacoustic sensing and the availability of radiated-noise datasets. Classical methods relied on handcrafted descriptors, such as cepstral coefficients, MFCCs, and narrow-band energy statistics, combined with classifiers such as k-nearest neighbors, support vector machines, and Gaussian mixture models. However, these approaches depend on prior feature design and are sensitive to noise, propagation variability, and sensor gain fluctuations.

Recent work has shifted toward deep learning methods applied to time-frequency representations. According to Domingos et al. (2022), *log-mel* spectrograms and convolutional neural networks have become common choices for underwater target and radiated-noise classification. Recent surveys further emphasize the importance of public datasets, data augmentation, environmental robustness, and stronger generalization protocols Luo et al. (2023); Hummel et al. (2024); Aslam et al. (2024). In this context, *DeepShip* has become a relevant benchmark for vessel-noise classification Irfan et al. (2021).

Recent architectures have explored complementary representations, attention mechanisms, and end-to-end learning. Tian et al. Tian et al. (2023) proposed a joint model combining waveform and time-frequency branches. Yang et al. Yang et al. (2024) used one-dimensional convolutions with Transformer blocks to capture temporal dependencies. Dong et al. Dong et al. (2025) introduced a cross-attention Transformer combining LOFAR, Mel, and wavelet-based representations. Chen et al. Chen et al. (2025) addressed multi-target recognition using channel-attention mechanisms.

Despite these advances, underwater engine recognition remains challenging due to background noise, spectral overlap among propulsion states, and limited annotated data. This work addresses these issues through a *multitask learning* formulation that jointly estimates the active engine and its operating state from short audio segments. By exploiting the relationship between both tasks, the proposed framework aims to improve file-level decisions for passive monitoring, acoustic warfare support, and maritime situational awareness.

3. MATERIALS AND METHODS

3.1 Wolfset dataset and single-engine subset

The *Wolfset* dataset Lobo et al. (2025) contains underwater acoustic recordings in WAV format sampled at 44.1 kHz. Each file is annotated by a five-digit intensity code that identifies the active engine or engines and their corresponding operating states. The dataset includes four combustion engines (M1–M4) and one electric motor (M5). In this work, M1–M5 denote only engine identities, whereas V1–V7 denote model variants.

This study focuses exclusively on the single-engine subset, aiming to establish a controlled benchmark for multitask recognition before addressing multi-engine mixtures. The full corpus also includes multi-engine combinations, approximately 15% of the data, and transient events such as *Air*, *Water bucket*, *Hammer/Mallet strike*, and *Shot*.

The single-engine split follows an 80/20 file-based partition. The training set contains 88 files, distributed as M1: 42, M2: 12, M3: 13, M4: 11, and M5: 10. The test set contains 21 files, distributed as M1: 10, M2: 3, M3: 3, M4: 3, and M5: 2. After segmentation, this yields approximately 6,300 segments, with 5,040 for training and 1,260 for testing. The test set includes five state categories: absent/disconnected, forward idle engaged, idle disengaged, medium forward, and fast or reverse.

3.2 Signals, windowing, and spectrograms

Audio signals are converted to mono and sampled at 44.1 kHz. Each file is divided into 0.5 s windows with 0.25 s hop, corresponding to 50% overlap, with zero-padding applied when necessary. For each segment, a *log-mel* spectrogram is computed using

$$(n_{\text{mels}}, n_{\text{fft}}, \text{hop}) = (64, 1024, 256),$$

with frequency range from 20 Hz to 8 kHz. All spectrograms are normalized using *z-score* normalization.

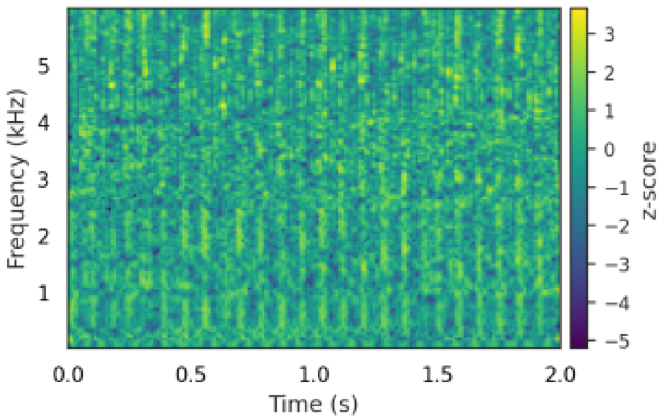


Fig. 1. Representative log-mel spectrogram, z-score normalized, for a combustion engine M1 under idle disengaged operation.

3.3 Multitask learning formulation

We formulate the problem as two correlated tasks: engine identification and engine-conditioned state recognition. Given an input segment x, a shared backbone ϕ produces a latent representation $z = \phi(x)$, which is passed to an engine classifier and to engine-specific state heads:

$$\begin{aligned} g_{\text{eng}}(z) &\in \mathbb{R}^5, \\ g_{\text{st}}^{(k)}(z) &\in \mathbb{R}^{C_k}. \end{aligned} \tag{1}$$

The total loss combines engine and state recognition:

$$\mathcal{L} = \lambda_{\text{eng}}\, \mathrm{CE}\big(g_{\text{eng}}(z), y_{\text{eng}}\big) + \lambda_{\text{st}}\, \mathrm{Focal}\big(g_{\text{st}}^{(k)}(z), y_{\text{st}}\big). \tag{2}$$

Unless otherwise stated, $\lambda_{\text{eng}} = \lambda_{\text{st}} = 1$. Engine recognition uses class-weighted cross-entropy, while state recognition uses *Focal Loss* with $\gamma = 2$. Only the state head associated with the ground-truth engine is optimized for each sample.

3.4 Backbone models

The backbone is the shared feature extractor that maps each spectrogram to a latent representation used by the engine and state heads. Four architectures are evaluated: BiLSTM with attention, GRU with attention, temporal Transformer, and ResNet-50.

GRU with attention The baseline recurrent model uses two bidirectional GRU layers followed by a mean-attention pooling mechanism, dropout, and the multitask head.

GRU with sub-spectrograms The proposed variant partitions the 64-bin mel spectrum into n_{bands} contiguous sub-bands. Each sub-band is processed by a shared two-layer bidirectional GRU and an attention module. The resulting band representations are concatenated, fused by a linear layer, and passed to the multitask head. The evaluated variants use $n_{\text{bands}} \in \{6, 8, 10, 12\}$.

BiLSTM with attention The BiLSTM baseline replaces the GRU encoder with two bidirectional LSTM layers while preserving the same attention, dropout, and multitask-head structure.

Transformer and ResNet-50 The temporal Transformer applies multi-head self-attention to the time–mel sequence with positional encoding and sequence pooling. The ResNet-50 baseline treats the spectrogram as an image, resizes it to 224 × 224, replicates it across three channels, and uses global average pooling to obtain the latent representation. Both models use the same multitask head for fair comparison.

Based on comparative results, we **propose a sub-spectrogram model** derived from the **GRU+attention** variant that performed best. The design partitions the *log-mel* spectrum into multiple contiguous bands and applies shared temporal encoding, as illustrated in Fig. 3.

The goal is to maximize information use in *log-mel* spectrograms. Processing the full spectrum may under-exploit local frequency patterns; sub-band factorization can extract more discriminative features. While Zhang et al. (2021) originally explored sub-spectrograms with CNNs, here we extend the principle to **recurrent encoders**: each sub-spectrogram is processed by a *shared* bidirectional GRU with per-band attention. The resulting vectors are fused before the multitask head, assessing whether Mel spectrogram decomposition broadens coverage of relevant patterns without unnecessary complexity.

Processing stages are: *split* → projection → temporal encoding → attention → fusion. Each band is first projected linearly ($w_b \to h$) into a common latent space, via **band-specific** projections. All pass through a **shared two-layer bidirectional GRU**, extracting temporal dynamics invariant to spectral crops and reducing parameter count. A **per-band attention** then yields $h_b \in \mathbb{R}^{2h}$.

Vectors $\{h_b\}$ are **concatenated** and fused linearly into $z \in \mathbb{R}^{2h}$, followed by dropout. The **multitask head** comprises (i) an *engine* classifier (5 classes, shared) and (ii) *state* classifiers (not shared), one per engine k, each with cardinality C_k. State logits are aligned via `NEG_INF` to ensure stable joint training.

3.5 Optimization and metrics

Training uses **Adam** with initial learning rate $\mathbf{2 \times 10^{-3}}$, chosen for numerical stability and adaptability in noisy-gradient regimes. A **ReduceLROnPlateau** scheduler (factor 0.5, patience 2) reduces LR after two epochs without validation improvement. Batches of size **32** are shuffled each epoch, with at most **50 epochs** and early stopping based on mean F1 stability.

We monitor, per epoch:

- **Loss** for *train* and *validation*, as the weighted sum of engine and state losses;
- **Accuracy** and **macro-F1** for each task on train and test.

Model selection uses the **mean F1 across tasks**:

$$\text{MeanScore} = \tfrac{1}{2}\left(F1_{\text{engine}} + F1_{\text{state}}\right).$$

This balances performance across the easier engine task and the harder state task. We adopt **macro-F1**—the unweighted mean across classes—given imbalances in engines and states; it penalizes skew and reflects minority-class performance (e.g., *state 4* with only 60 segments).

3.6 Segment-level and file-level evaluation

We report accuracy and macro-F1 at segment and file levels. Segment-level metrics are computed on individual windows and quantify local discrimination under short-term spectro-temporal variability. File-level decisions are obtained by averaging segment logits, then applying softmax and `argmax`. We adopt mean-logit aggregation as the default score-fusion rule because it is simple and less sensitive to isolated high-confidence segment errors than hard voting. Macro-F1 is used throughout because of class imbalance, especially for rare state classes. In the revised evaluation, file-level aggregation is also compared with mean probabilities and majority vote.

3.7 Light Data Augmentation

To reduce overfitting and improve robustness to acoustic variability, we applied a light data augmentation scheme during training. Three independent transformations were used, each with probability $p = 0.5$.

First, random gain was applied in the time domain:

$$g_{\text{dB}} \sim \mathcal{U}(-3, 3), \qquad y'[n] = y[n]\, 10^{g_{\text{dB}}/20}.$$

This simulates moderate variations in acoustic intensity and sensor gain.

Second, low-amplitude Gaussian noise was added:

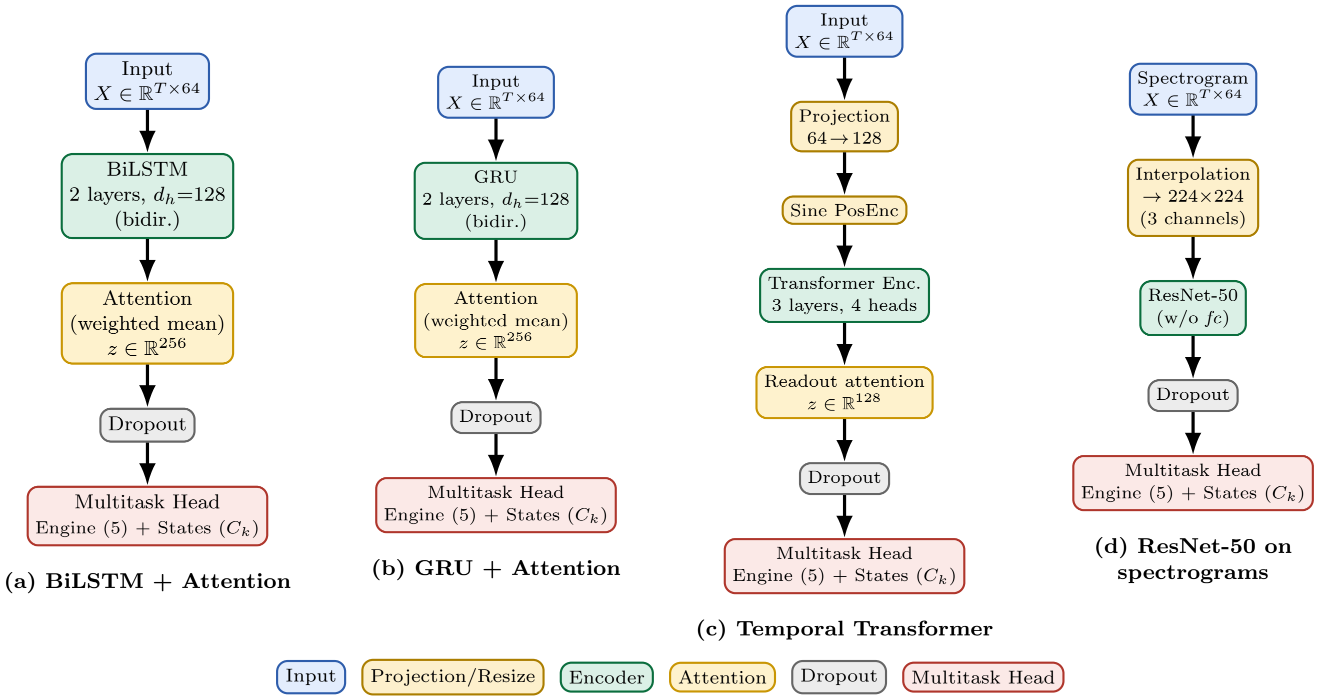
$$\begin{aligned} y''[n] &= y'[n] + r[n], \\ r[n] &\sim \mathcal{N}\big(0, (\alpha\sigma_y)^2\big), \\ \alpha &\sim \mathcal{U}(0, 0.01). \end{aligned} \tag{3}$$

This increases tolerance to background noise and small acoustic perturbations.

Third, after log-mel conversion and normalization, light *SpecAugment* was applied using random time and frequency masks covering up to 10% of the corresponding

Table 1. Core hyperparameters of the backbones.

Backbone	Block	Layers	Hidden Dim	n_{bands}	Attention	Dropout
`V1/V7`	Bi-GRU	2	96 (per direction)	—	MeanAttention	0.20
`V2`	Bi-GRU/band	2	96 (per direction)	variable	MeanAttention (per band)	0.15
`V3`	Bi-GRU/band	2	96	6	MeanAttention (per band)	0.15
`V4`	Bi-GRU/band	2	96	8	MeanAttention (per band)	0.15
`V5`	Bi-GRU/band	2	96	10	MeanAttention (per band)	0.15
`V6`	Bi-GRU/band	2	96	12	MeanAttention (per band)	0.15
BiLSTM + attention	Bi-LSTM	2	—	—	Attention	—
Transformer	self-attn	—	—	—	*CLS*/pool	—
ResNet-50	2D conv.	—	—	—	*GAP*/pool	—



(a) BiLSTM + Attention **(b) GRU + Attention** **(c) Temporal Transformer** **(d) ResNet-50 on spectrograms**

Fig. 2. Models with **vertical** flow (side-by-side) and color-coded blocks: input (blue), projection/resize (orange), encoder (green), attention/pooling (yellow), dropout (gray), and multitask head (red). Left to right: (a) BiLSTM + Attention; (b) GRU + Attention; (c) Temporal Transformer; (d) ResNet-50.

axis. This encourages robustness to local spectral occlusions and temporal variability.

The transformations were applied independently and with randomized order, increasing statistical diversity without changing the number of training samples or introducing strong distortions.

3.8 Sub-spectrogram architecture

In the proposed sub-spectrogram model, the 64-bin mel spectrogram is divided into B contiguous frequency bands. Each band is projected to a latent space and processed by a shared two-layer bidirectional GRU with hidden size $h = 128$. A per-band attention module produces band-specific embeddings h_b, which are concatenated and fused into a $\mathbb{R}^{2h}$ representation before dropout and the multitask head.

We evaluate $B \in \{4, 6, 8, 10, 12, 14\}$. This design allows the model to learn frequency-specific cues for state recognition, while sharing the GRU across bands controls the number of parameters and supports generalization.

4. RESULTS

4.1 Main backbone comparison

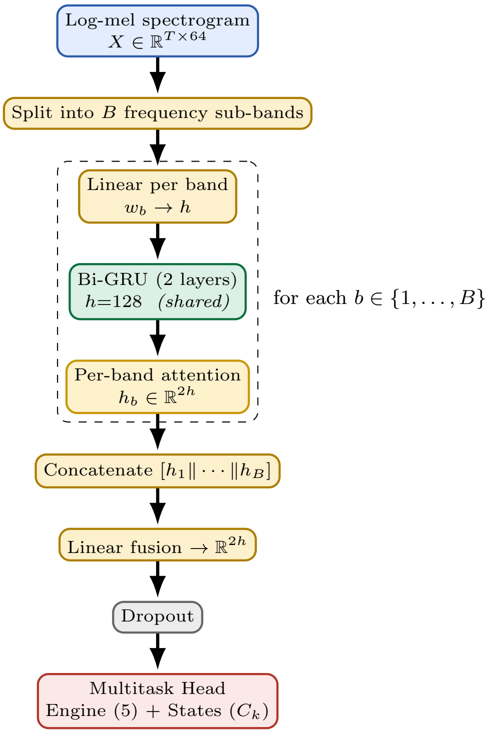
Table 2 compares BiLSTM+Attention, GRU+Attention, Transformer, and ResNet-50 on the single-engine subset using $(T, H) = (0.5, 0.25)$ s. At segment level, GRU achieves the best engine macro-F1 of 0.97, while LSTM and Transformer obtain the highest state macro-F1 of 0.57. At file level, LSTM, GRU, and Transformer reach engine macro-F1 of 1.00, while Transformer obtains the best state macro-F1 of 0.68, followed by GRU with 0.66.

Most errors occur between adjacent operating states and between acoustically similar engines, especially M3 and M4 under comparable loads. File-level aggregation reduces these confusions, particularly for engine recognition.

The sub-spectrogram variants are compared in Table 3. Performance improves up to $B = 8$ in V4, which achieves the best overall mean score of 0.800 and the best file-level state F1 of 0.74. Larger values of B provide no

Table 2. Macro-F1 at segment and file levels on the single-engine subset, $(T, H) = (0.5, 0.25)$ s.

Model	$\mathbf{F1_{seg}}$ Engine	$\mathbf{F1_{seg}}$ State	$\mathbf{F1_{file}}$ Engine	$\mathbf{F1_{file}}$ State
LSTM BiLSTM+Att	0.93	0.57	1.00	0.53
GRU+Attention	**0.97**	0.56	**1.00**	0.66
Transformer	0.93	**0.57**	**1.00**	**0.68**
ResNet-50	0.87	0.55	0.90	0.66



(e) GRU with sub-spectrograms (split → per-band proj. → shared GRU → per-band attention → concat/fusion)

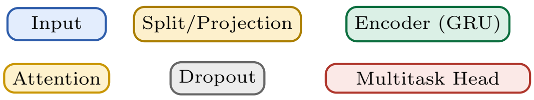


Fig. 3. Sub-spectrogram model: the mel-spectrogram is partitioned into B bands; each band goes through a linear projection and a **shared** bi-GRU; per-band attention yields h_b, which are concatenated and fused linearly before dropout and the multitask head.

consistent improvement, suggesting saturation in the sub-band decomposition.

4.2 Experiment V7: Evaluation without data augmentation

To assess the impact of augmentation, V7 repeats the recurrent sub-spectrogram configuration without random gain, Gaussian noise, or *SpecAugment*. Results are shown in Table 4.

Without augmentation, engine recognition remains relatively stable, but state recognition degrades substantially. Compared with V4, file-level F1 decreases from 1.00 to 0.90 for engine recognition and from 0.74 to 0.47 for state recognition. This confirms that light augmentation is essential for improving generalization, especially for operating-state classification, where classes are more imbalanced and acoustically similar.

4.3 Computational performance

Experiments were run on a `Tesla T4` GPU (16 GB). Table 5 reports end-to-end inference time per complete recording under the adopted windowing (T=0.5 s, H=0.25 s) and evaluation batching, together with peak memory. The observed 83–113 s should therefore be interpreted as per-file processing time rather than per-segment latency, supporting batch or near-online monitoring but not strict real-time deployment.

Table 5. End-to-end per-file inference time and peak memory (Tesla T4 GPU).

Model	Time (s)	Memory (MB)
V1	88.4	210.6
V2	99.6	213.3
V3	95.8	215.1
V4	96.1	216.1
V5	95.6	218.1
V6	113.0	220.9
V7	83.1	187.4

Practical conclusions. (i) Variant V4 (n_{bands}=8) provides the best overall trade-off between spectral granularity and training stability. (ii) Engine identification reaches file-level F1 of 1.00 in several variants, indicating that moderate sub-band partitioning is sufficient for the single-engine subset. (iii) State recognition benefits more strongly from sub-band factorization and temporal aggregation. (iv) Gains beyond eight sub-bands are marginal. (v) These findings are restricted to the single-engine subset of *Wolfset* and should be interpreted as evidence for controlled monitoring, not for general multi-engine deployment.

In summary, V7 demonstrates that, for the *Wolfset* dataset, the combination of random gain, white noise, and *SpecAugment* is decisive for training stability and for faithful file-level state estimates.

5. CONCLUSION

This work presented a reproducible multitask pipeline for recognizing the active underwater engine and its operating state from short hydroacoustic segments. The proposed system combines *log-mel* spectrograms, light data augmentation, a shared backbone, and engine-conditioned state heads, and can be viewed as a perception module for monitoring and higher-level decision-support systems.

Within the single-engine subset of *Wolfset*, the GRU-based sub-spectrogram variant with n_{bands}=8 achieved the best overall trade-off between accuracy and computational cost. These results are encouraging for controlled single-source monitoring, but they do not yet establish performance

Table 3. Macro-F1 by level and test mean score for models V1–V6.

Model	$F1_{seg}$ Engine	$F1_{seg}$ State	$F1_{file}$ Engine	$F1_{file}$ State	Mean Score
V1	0.97	0.66	1.00	0.70	0.779
V2	0.94	0.60	0.93	0.70	0.782
V3	0.93	0.59	1.00	0.68	0.771
V4	0.94	**0.66**	1.00	**0.74**	**0.800**
V5	0.92	0.62	0.93	0.75	0.785
V6	0.92	0.61	0.95	0.70	0.776

Table 4. Performance of V7 without data augmentation.

Task	Segment Level				File Level			
	Precision	Recall	F1	Accuracy	Precision	Recall	F1	Accuracy
Engine	0.94	0.89	0.90	0.91	0.97	0.87	0.90	0.90
State	0.62	0.46	0.49	0.67	0.67	0.43	0.47	0.67

in simultaneous multi-engine scenes or strict real-time settings.

Future research directions include: (i) extending the approach to multi-*label* conditions with multiple simultaneously active engines; (ii) incorporating calibration and uncertainty estimation mechanisms; (iii) integrating audio–visual modules for platform recognition; and (iv) adapting the architecture to embedded, power-constrained environments through quantization and TinyML for edge deployment.

ACKNOWLEDGEMENTS

The authors express their gratitude to researchers **Nuno Pessanha Santos**, **Ricardo Moura**, and **Victor Lobo** for publicly releasing the dataset *Wolfset: A High-Quality Underwater Acoustic Dataset for Algorithm Development and Analysis*, which served as the experimental foundation for this study.

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